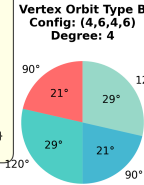
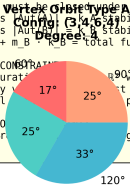


2-Uniform Tessellations: Constraint Equations and Verification

2-UNIFORM TESSELLATION EXAMPLE: Two Vertex Orbits

Constraint System for validity:

1. ANGLE CONSTRAINT at Type A: $60^\circ + 90^\circ + 120^\circ + 90^\circ = 360^\circ$ ✓
2. ANGLE CONSTRAINT at Type B: $90^\circ + 120^\circ + 90^\circ + 120^\circ = 360^\circ$ ✓
3. TOPOLOGICAL CONSTRAINT (Euler):
 - For fundamental domain with V vertices, E edges, F faces:
 $V - E + F = 2$ (for torus: $V - E + F = 0$)
 - Lattice version: $\sum_i m_i = \text{constant}$ (consistency)
4. ORBIT CONSTRAINT:
 - Each orbit must be closed under automorphisms
 - Orbit A has $|Aut(A)| = k_A$ stabilizers
 - Orbit B has $|Aut(B)| = k_B$ stabilizers
 - $m_A \cdot k_A + m_B \cdot k_B = \text{total fundamental domain size}$
5. PERIODICITY CONSTRAINT:
 - The configuration must be without gaps or overlaps
 - Periodicity must generate primitive lattice
 - Fundamental domain must have positive area
6. WALLPAPER GROUP:
 - Symmetry group: {id, g, p2mm, p2mg, p2gg, c2mm, p4, p4m, p4g, p3, p3m1, p3m, p6, p6m}



Constraint	Type A	Type B	Status
Angle Sum = 360°	360°	360°	✓ PASS
Vertex Degree	4	4	✓ CONSISTENT
Polygon Types	3	2	✓ DISTINCT
Orbit Closure	Closed	Closed	✓ VALID
Periodicity	Tileable	Tileable	✓ VALID
Wallpaper Group	p4m	p4m	✓ COMPATIBLE